Physics Is Already There: Rethinking Physical Injection in Latent World Models

Anonymous submission

Abstract

A world model predicts how an environment evolves in response to actions by rolling an internal representation forward, and is expected to respect physical regularities such as how objects move and collide. We study JEPA-style latent world models and find a puzzle: trained with no physics supervision, such a model already recovers object position well from its representation and predicts one step almost exactly, yet its multi-step rollouts drift away from the very dynamics it encodes. The popular fix is to inject physical structure into the latent representation. We test this exhaustively—ten injection variants spanning hard and soft constraints on both physical quantities and physical laws, with labeled and label-free supervision—across three controlled physics domains and two realistic simulation benchmarks, and find injection fails in 29 of 30 cases; even supplying the *correct* physical constraint makes prediction worse. We trace this to a single mechanism: the physical state is already encoded across the model's internal representation, so an injected copy is superfluous—prediction reads the state from the internal representation and bypasses the injected one. What the model lacks is not the object's physical state but the ability to evolve that state over long horizons: fixing the training procedure alone, adding no physics, cuts rollout error 2.2–8.3$\times$ on identical code and metrics. To help a latent world model, injected structure should supply what the representation is missing—*how the state evolves, not what the state is*.

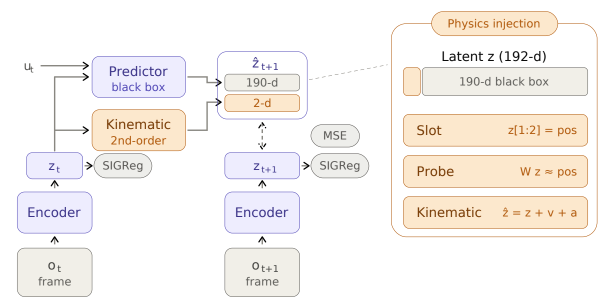

Figure 1: Physics injection into the LeWM shared-latent world model. *Amber* marks everything injection adds to the backbone; the rest is unmodified LeWM. The encoder maps each frame to a 192-d latent; a black-box transformer predicts the full latent from the latent history and action u_t, and the free-rollout objective aligns the predicted $\hat{z}_{t+1}$ to the encoded next latent by MSE (SIGReg prevents collapse). Physics enters a 2-d position *slot* carved from the 192 dimensions (right panel): a hard *slot* pinning $z[1{:}2]$ to position, a supervised *probe* reading position off the latent, or a *kinematic* head evolving the slot by a second-order rule—while the other 190 "black-box" dimensions stay free. This 2-vs-190 split is the crux: prediction can route through the black box and bypass any injected slot.

1 Introduction

A *world model* rolls an internal representation forward to predict how an environment evolves under an agent's actions—a simulator to plan with instead of acting in the world (Ha and Schmidhuber 2018; LeCun 2022). For a physical environment, it should respect how objects move, fall, and collide. JEPA-style latent world models (Maes 2025; Balestriero and LeCun 2025) present a puzzle here. Trained on PhyWorld (Kang et al. 2025) with no physics supervision, such a model already recovers object position from its representation by linear probing (correlation up to 0.96) and predicts one step almost exactly (latent cosine 0.98–0.99)—yet rolled out autoregressively, its predictions drift away from the very dynamics it encodes: on collision the latent cosine falls to 0.24 by horizon 28, and out-of-distribution (OOD) physics—unseen radii, masses, or velocities—multiplies rollout error several-fold. This indicates the state is already present in the representation; the model simply fails to *evolve* it.

The popular fix is to *inject physical structure into the latent* (Figure 1): a dedicated slot pinned to the physical state, a supervised probe that reads it out, or a hard-wired dynamics rule. Such injection has real successes in adjacent model classes—architectures that route prediction through a low-dimensional physical state by construction (Mao, Umasudhan, and Ruchkin 2024; Peper et al. 2025), deep supervision where the supervised quantities occupy a large fraction of a compact state latent (Zahorodnii 2025), and a decomposed JEPA latent supervised on low-dimensional numerical time series (Nie et al. 2026). Scaling data is not the answer either: video generators trained at scale still fail OOD physics, memorizing cases rather than laws (Kang et al. 2025). But for a *shared, high-dimensional* latent—where prediction runs end-to-end through one representation—injection has only been tried in isolated, incomparable settings, and never against the confound that dominates the outcome: the training protocol. Whether physical structure adds anything, once the protocol is done right, has never been tested systematically.

We answer this with a systematic, controlled anatomy on a compact JEPA world model. We cross ten injection variants—spanning hard and soft constraints, targeting the physical *state* (what it is) or its *evolution* (how it changes), under labeled and label-free supervision—with two training regimes and three controlled physics domains, and validate the surviving conclusions on two realistic-simulation benchmarks (Bear et al. 2021; Tung et al. 2023). Verdicts rest on scale-aware prediction error (latent nMSE and pixel PSNR), never on probe correlation or cosine—diagnostics that rise mechanically with the injected loss and can manufacture apparent gains; all headline comparisons are seed-controlled (§4.1).

The structure does not help. Injection degrades or fails to improve prediction in 29 of 30 mechanism–domain cells, and the failures are pointed. Supplying the *correct* physical form—the exact gravitational law $a{=}g$—hurts as much as an unconstrained dynamics head ($1.57\times$ the baseline error on uniform motion). Co-training the structure from scratch, rather than grafting it onto a pretrained encoder, hurts *more*, not less ($+0.558$ vs. $+0.035$ nMSE)—closing the defense that physics must be baked in during pretraining. The lone seed-robust gain reverses sign across domains: a diagnostic signature, not a usable prior.

Why should the *correct* physics hurt? Because the physical state is *already there*. A pinned slot occupies 2 of the 192 latent dimensions, but the other 190 redundantly encode the same position—decoding it nearly as accurately as the full latent—so prediction routes around the injected slot and reads the copy it already carries. This account makes a falsifiable prediction: if the slot's $\sim1\%$ share of the prediction loss is what starves it, up-weighting the slot should remove the harm. It does not—re-weighting to saturation recovers only *parity*, never a net gain, because no loss weight closes a bypass wired into the architecture. The state is *decodable, but not load-bearing*.

What the representation lacks is not the state but the ability to *evolve* it over long horizons—and that gap is fixable with no physics at all. Repairing the training protocol alone—free rollout, with the horizon matched to each domain's dynamics—cuts rollout error 2.2–8.3$\times$ on the identical code, metrics, and seeds that judged every injection cell, pinning the failure on the injected structure rather than the pipeline or the yardstick. The lesson is constructive: to help a shared-latent world model, an inductive bias must supply what the representation is missing—how the state evolves—not re-encode what it already holds. An architecture that makes the physical state the *sole* pathway prediction can take does close the bypass; but ported faithfully to our benchmark, it collapses once an OOD shift reaches its encoder (ρ 0.33 vs. 0.89)—necessary, but not sufficient. Every finding above replicates on a second, frozen-DINOv2 backbone.

This paper contributes:

- the first systematic, seed-controlled anatomy of physical inductive biases in *visual* latent world models—ten variants $\times$ two training regimes $\times$ three domains, plus two realistic-simulation benchmarks—finding injection no better than baseline in 29 of 30 cells, and replicating on a frozen-DINOv2 backbone (§4.2, §4.5);
- a mechanistic account, *decodable but not load-bearing*, from a bypass probe, a loss-dominance and effective-dimensionality measurement, and a dose–response sweep; its architectural prediction is confirmed externally—an extrinsic port closes the bypass yet collapses under encoder-OOD shift: necessary, not sufficient (§4.3, §4.5);
- a controlled factorial showing the dominant variable is the training protocol, not physical structure: free rollout alone cuts rollout error 2.2–8.3$\times$ with no physics added, doubling as the positive control against implementation and metric artifacts (§4.4).

2 Related Work

Physically structured world models. A growing line argues that world models should carry explicit physical state: physically interpretable models route prediction through a dedicated low-dimensional state (Mao, Umasudhan, and Ruchkin 2024; Peper et al. 2025); deep supervision adds physical readouts to the loss and helps in a low-dimensional state latent, where the supervised quantities occupy $\sim38\%$ of the representation (Zahorodnii 2025); closest to ours, Phys-JEPA supervises a decomposed JEPA latent and reports gains (Nie et al. 2026)—but on numerical time series, not the visual latents we study, a distinction that reverses the conclusion (§4.3). Physics-informed losses (Raissi, Perdikaris, and Karniadakis 2019) and conservation-structured dynamics (Greydanus, Dzamba, and Yosinski 2019; Cranmer et al. 2020) constrain the model class; graph simulators over *explicit* particle state predict physics well (Battaglia et al. 2016; Sanchez-Gonzalez et al. 2020)—structure works when the physical state *is* the representation, not a small part of a shared one. We test the transferable core of these proposals inside a shared-latent *visual* JEPA model and find all of them unhelpful or harmful (§4.2); our extrinsic port (§4.5) attributes the gap to architecture, not to the physics.

Latent world models and physics benchmarks. JEPA-style models predict in representation space (Assran et al. 2023; Bardes et al. 2024; Balestriero and LeCun 2025; Maes 2025); DINO-WM plans over frozen features (Zhou et al. 2025; Oquab et al. 2024), the design we adopt as our cross-backbone control; Dreamer-line models train on imagined rollouts (Ha and Schmidhuber 2018; Hafner et al. 2025). PhyWorld (Kang et al. 2025) supplies controlled single-mechanism physics videos with explicit in-distribution ranges, showing that scaled video generators memorize cases rather than laws; Physion and Physion++

(Bear et al. 2021; Tung et al. 2023) add realistic—not camera-captured—simulation. We apply PhyWorld's OOD protocol to *latent* models rather than pixel-space generators.

Rollout training and exposure bias. Training predictors on their own outputs is a classic remedy for compounding error (Bengio et al. 2015; Venkatraman, Hebert, and Bagnell 2015; Goyal et al. 2016). We claim no novelty for free rollout; its role here is methodological (§3.4): the controlled factorial variable prior injection studies never controlled, and a positive control separating "structure does not help" from "the implementation or metrics are broken."

3 Method: A Design Space of Physical Injection

We answer the question of §1 by construction: design the ways physics can be injected into a shared latent, state what each design is supposed to achieve, and lay out the experiments that test the mechanism behind it.

3.1 Setting and Notation

We study LeWM (Maes 2025), a compact JEPA-style action-conditioned world model ($\sim$10M parameters): an encoder E_θ (ViT-tiny) maps frame o_t to a latent $z_t \in \mathbb{R}^D$ with D=192, and a causal-transformer predictor F_ϕ rolls latents forward from the history and action u_t, i.e. $\hat{z}_{t+1} = F_\phi(z_{\leq t}, u_t)$. Training minimizes a prediction loss, an anti-collapse regularizer (Balestriero and LeCun 2025), and—when physics is injected—an auxiliary physical loss:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \beta\,\text{SIGReg}(z) + \lambda\,\mathcal{L}_{\text{phys}}. \tag{1}$$

Here $\mathcal{L}_{\text{pred}} = \frac{1}{D}\sum_i w_i(\hat{z}_{t+1,i} - z_{t+1,i})^2$ is a per-dimension weighted prediction error. By default w_i=1; *re-weighting* raises w_i to w on the 2 slot dimensions (w swept $1 \to 300$ in §3.3), while λ scales the soft auxiliary losses (probe, consistency, label-free priors). Throughout, w denotes the slot's prediction-loss weight and λ an auxiliary-loss weight—two distinct dials. *Teacher forcing* trains F_ϕ on one-step transitions from encoded ground-truth context; *free rollout* feeds predictions back for H autoregressive steps (H=8 unless stated) and supervises the whole rollout. The kinematic head (Family 3 below) replaces the slot's transition by a second-order update

$$\hat{z}^{[0:2]}_{t+1} = z^{[0:2]}_t + v_t\,\Delta t + \tfrac{1}{2}\,a\,\Delta t^2, \tag{2}$$

where v_t is the finite-difference slot velocity and the acceleration a is either a learned $\text{MLP}(z_t, u_t)$ or a single learnable constant—the strict a=g head. Unless stated otherwise the encoder is initialized from the public PushT checkpoint; §4.2 additionally studies from-scratch training.

3.2 Injection Design Space: Five Families, Ten Variants

We span the design space along three orthogonal axes—*hard vs. soft* constraints on the physical quantity, pinning *what the motion state is* vs. *how it evolves*, and *labeled vs. label-free* supervision—so that any "try a different encoding" objection lands on an axis we have measured. Table 1 lists the ten variants with the design rationale of each.

Variant	What it does, and what it tests
Family 1: fixed slot — hard constraint, pins the state, labeled	
slot (structpos)	pin dims $[0{:}2]$ to the true position (the object's x, y coordinates; a fixed index range, not an attention slot (Locatello et al. 2020)): the most direct state injection; tests whether a hard constraint helps
+reweight	raise the slot's share of the prediction loss (w=30, Eq. 1): if failure is due to the slot's $\sim$1% gradient share, this should rescue it
+velocity	encode velocity too: tests whether it matters *which* quantity is injected, not just that something is
Family 2: deep-supervision probe — soft constraint, pins the state, labeled	
probe	linear head reads position from the latent: replicates the deep-supervision recipe (Zahorodnii 2025) in a high-dimensional visual latent
+slot	soft + hard combination: tests complementarity
Family 3: kinematic head — hard constraint, pins the evolution, labeled	
free MLP	attach $z{+}v{+}a$ update with learned a: upgrade from pinning state to pinning *evolution*
strict a=g	exact gravitational form, one learnable constant: closes the "your physics was wrong" objection
Family 4: consistency — soft constraint, pins the evolution, labeled	
consistency	finite-difference velocity of the rolled-out positions must match ground truth; assumes no form for a, designed for collision impulses
Family 5: label-free prior — soft constraint, pins the evolution, label-free	
label-free	slot must evolve as smooth second-order dynamics, no ground truth: the only injection available for raw video
grounded	same structure + true labels: isolates the label variable

Table 1: Ten injection variants in five families, spanning three axes (read off each family's header). **Constraint**—*hard*: a latent dimension is forced to equal the physical quantity; *soft*: an auxiliary loss only encourages it. **Target**—the *state* (what the motion is: position, velocity) vs. its *evolution* (how it changes: a dynamics rule). **Supervision**—*labeled* uses ground-truth physics, *label-free* does not. The three axes span the ways to inject physics into a shared latent, so any alternative encoding falls on an axis we test. All variants use the strongest training protocol (§3.4).

Orthogonally, we cross injection with the *training regime*: fine-tuning a pretrained encoder vs. from-scratch co-training; with injection on/off this gives a 2×2 per domain. This closes the remaining escape hatch—"structure must be baked in during pretraining, as extrinsic architectures do."

3.3 Mechanistic Hypothesis and Its Tests

Hypothesis (the load-bearing problem). The physical slot occupies 2 of 192 dimensions and receives $\sim$1% of the prediction gradient, while the black-box channel *redundantly* encodes the same position—so prediction can route around any injected slot. Because the latent already carries the injected state, injection adds no new signal—it only consumes capacity and gradient share. We test this account in three ways, each admitting a specific outcome that would falsify it:

1. **Is the physical constraint injected?** Split the latent into the 2 slot dimensions and the 190 black-box dimensions and decode position from each separately, with a random-2-dimension control (Hewitt and Liang 2019). If the black box alone cannot decode position, the redundancy claim is false.
2. **Is the constraint acting at all?** Two measurements: how large the weighted physical auxiliary loss is relative to the prediction loss (a proxy for gradient magnitude), and whether the encoder's effective dimensionality (participation ratio) changes. If neither moves, the constraint never took effect and the failure is a bug. If the auxiliary term dominates and dimensionality collapses, the constraint is acting—and competing with prediction.
3. **Does re-weighting the physical loss help?** Sweep the slot weight w of Eq. 1 from 1 to 300. If harm does not respond systematically to the weight, the gradient-share mechanism is wrong; if it responds but weighting to saturation still cannot rescue every domain, demonstrating that insufficient physical loss is not the root cause.

3.4 The Training-Protocol Control: Free Rollout

Free rollout serves two methodological roles; neither is a novelty claim (it is scheduled sampling, Bengio et al. 2015).

Controlled factorial variable. Physical structure has never been compared against the training protocol on the same baseline; prior studies treat the protocol as background. We set the protocol first—replacing teacher forcing with free rollout (§3.1)—and then ask whether structure adds incremental value. Teacher forcing hides compounding error at training time (*exposure bias*): the model never sees its own mistakes, then must consume them at deployment. Free rollout exposes them during training, injecting no physics whatsoever. We read the resulting gain as more than robustness to compounding error: supervising an H-step autoregressive rollout turns training itself into *long-horizon dynamics simulation*. Under teacher forcing the predictor is graded only on reproducing local one-step transitions; under free rollout it is graded on *evolving* the state over the full horizon, so the gradient rewards a self-consistent multi-step dynamics rather than a one-step map—precisely the capability the latent is missing (§1). On this reading, free rollout is an evolution-targeted training signal that requires no physics labels: it trains *how the state evolves* without touching *what the state is*. A second protocol variable, the rollout horizon H, must *match dynamics complexity*: domains with late causal events (collision; realistic-simulation scenes) need horizons long enough to contain them.

Positive control. A negative result invites two deflations—"your implementation is broken" or "your metrics cannot detect improvement." A protocol change that produces large gains on identical code and identical metrics rules out both at once, pinning the failure on the injected structure itself.

3.5 External and Cross-Backbone Controls

Two controls probe our account from outside the LeWM backbone: one tests whether an architecture that *closes* the bypass succeeds where injection fails, the other whether the conclusions survive a different encoder.

Extrinsic port. We faithfully port the physically interpretable world model (PIWM) of Mao, Umasudhan, and Ruchkin (2024)—a three-stage pipeline (VAE encoder, 128-D; a supervised extractor to a low-dimensional physical state; known-equation dynamics with learnable constants)—onto the same PhyWorld domains and OOD protocol. The port tests our mechanism's architectural prediction from the outside: extrinsic designs make the physical state the sole pathway prediction must route through, closing the bypass by construction.

Cross-backbone control. To test that the conclusions are not artifacts of one encoder, we replicate the core protocol on a second JEPA-family instantiation in the style of DINO-WM (Zhou et al. 2025): a *frozen* DINOv2-small encoder (Oquab et al. 2024)—generic self-supervised features that have never seen PhyWorld—with the trainable projector serving as adapter and the identical predictor stack, losses, seeds, and evaluation. The variant differs from LeWM in backbone provenance, dimensionality, and, crucially, in whether the injected losses can reshape the encoder at all.

4 Experiments

4.1 Setup

Model. The compact, fully controlled backbone of §3.1 (LeWM: E_θ, an MLP projector, an action encoder, and a 6-layer causal-transformer F_ϕ) allows the complete factorial anatomy ($\sim$200 training runs), with headline configurations repeated over three seeds (3072/1234/42); remaining cells are single-seed and marked as such.

Datasets and OOD protocol. All benchmarks are *visual* world-modeling tasks (RGB video frames), not the numerical/tabular sequences of prior physics-injection work (§2); injected state is thus always a small fraction of a high-dimensional visual latent. *PhyWorld* (Kang et al. 2025) provides three single-mechanism synthetic domains—*uniform motion*, *parabola*, *collision* (zero/constant/impulsive acceleration)—each with $1{,}000$ ID trajectories (32 frames, 224×224; radius/mass $\in [0.7, 1.5]$, velocity $\in [1, 4]$), evaluated on four partitions (ID, r/m-OOD, v-OOD, both-OOD); its single-variable OOD ranges isolate physical extrapolation from data-scale confounds. Two *realistic-simulation* benchmarks (rendered, not camera video) add realism: *Physion* (Bear et al. 2021) for zero-shot object-contact-prediction (OCP) transfer, and *Physion++* (Tung et al. 2023)—ten physical-property scenarios (mass/friction/bounce/deformation)—for direct training with rollout to horizon 64.

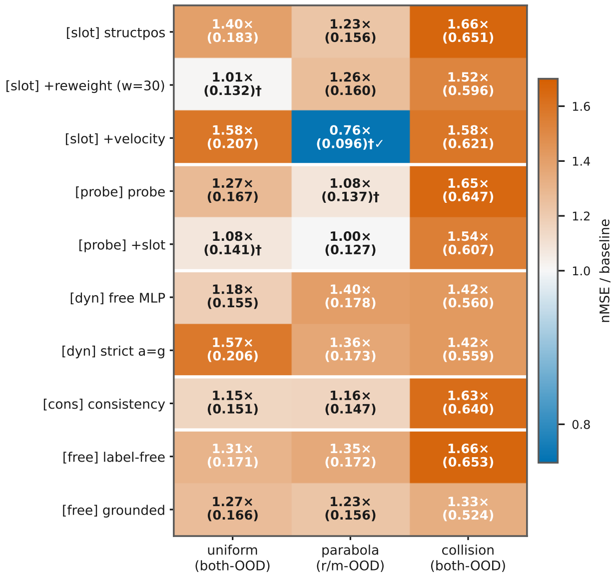

Figure 2: Every injection variant × domain against the free-rollout baseline (nMSE ratio; † = three-seed mean, others single-seed). Vermillion = worse, white = parity, blue = better. Injection is at or below baseline in 29 of 30 cells (25 worse, 4 parity); the collision column (1.33–1.66×) is uniformly worst; the sole real gain is velocity+slot on parabola (0.76×, †✓).

Metrics and decision rules. Verdicts use two scale-aware, *external* metrics—latent nMSE ($\|\hat{\mathbf{z}} - \mathbf{z}\|^2$ over target variance) and pixel PSNR of decoded rollouts—external because no injection variant optimizes them directly, and they agree in every case we examined. Latent cosine and linear-probe decodability ρ (Alain and Bengio 2016) are *diagnostics only*: both are duals of the auxiliary losses and rise mechanically when those are optimized, overstating physical competence (we measured cosine $1.50\times$ better while nMSE degraded to $0.21\times$). nMSE fails oppositely—its denominator degenerates when target variance collapses—so parabola is judged on its clean r/m-OOD partition, and all numbers are reported per partition and per horizon. Full pathology catalogue: Appendix E (supplementary). Splits are trajectory-level; three-seed results report mean±sample std.

4.2 Injection Fails in 29 of 30 Cells

Injection degrades or fails to improve prediction in 29 of 30 mechanism–domain cells (25 worse, 4 at parity; Figure 2). Every cell carries the raw nMSE alongside the ratio, so the figure doubles as the full result table (seed annotations in Appendix B (supplementary)); read family by family, it answers every design rationale of Table 1. *Hard state injection*: the slot faithfully absorbs position (its supervision loss falls $2.53 \to 0.015$; slot decodability rises $0.31 \to 0.96$), yet prediction worsens in all three domains ($1.40\times/1.23\times/1.66\times$)—the constraint acts, harmfully. *Soft injection*: the deep-supervision recipe, effective in a low-

Domain	Regime	phys. off	phys. on	Δ
uniform	scratch	0.192	0.750	**+0.558**
	fine-tune	0.131	0.166	+0.035
parabola (r/m-OOD)	scratch	0.343	0.375	+0.03
	fine-tune	0.124	0.198	+0.07
collision	scratch	0.538	0.635	+0.097
	fine-tune	0.393	0.525	+0.132

Table 2: Injection on/off × training regime, per domain (latent nMSE ↓ on the hardest clean split; injected structure: strict $a{=}g$ head + fixed slot). *Scratch* trains everything from random initialization; *fine-tune* starts the encoder from pre-trained weights. Δ is the cost of switching injection on within a regime: positive in all six regime–domain cells, and amplified from $+0.035$ to $+0.558$ on uniform when physics is baked in from the start.

dimensional state latent where supervised quantities occupy $\sim$38% of the representation (Zahorodnii 2025), does not transfer to a 192-D visual latent where they occupy 2%: all three domains degrade ($1.27\times$/parity-after-seeds/$1.65\times$), and combining soft with hard does not rescue ($1.54\times$ on collision). *Correct physics does not help*: the strict $a{=}g$ head—the ground-truth functional form—degrades uniform by $1.57\times$ and parabola from 0.127 to 0.173, no better than an unconstrained MLP (0.178). *The purpose-built design fails worst on its target*: the consistency loss, designed for collision impulses, is among the worst exactly there ($1.63\times$). *Labels are not the issue*: the label-free prior and its perfectly labeled counterpart fail in the same direction (collision $1.66\times$ vs. $1.33\times$)—the failure is architectural, not informational.

The sole exception is the mechanism's signature, not a usable prior: adding *velocity* to the re-weighted slot helps on parabola (r/m-OOD $0.122{\pm}0.007 \to 0.096{\pm}0.007$, every seed lower) because there velocity is an extrapolable driving variable ($a{=}g$ makes it linear in time); the same encoding is the *worst* variant on uniform ($+0.076$, velocity is a redundant constant) and clearly harmful on collision ($+0.025$ over the reweighted slot; velocity jumps discontinuously). A prior that requires knowing the domain's dynamics in advance, and reverses sign across domains, is a diagnostic, not a method—and its best case (-0.026 nMSE) is an order of magnitude below the protocol lever of §4.4.

From-scratch co-training hurts more. The remaining defense—structure must be baked in during pretraining—fails in the strongest way (Table 2): the physics penalty on uniform grows from $+0.035$ (fine-tuned) to $+0.558$ (from-scratch); all six regime–domain cells are positive. The cleaner the baseline, the larger the harm.

Two corroborations independent of PhyWorld nMSE. First, *zero-shot transfer* (different dataset, metric, and task): on Physion OCP, no synthetic-trained configuration exceeds the random-encoder prior (0.607 mean AUC); the physics-structured variant is the *worst* of all (0.551), below free rollout (0.603)—"more structure, worse transfer" (Appendix E

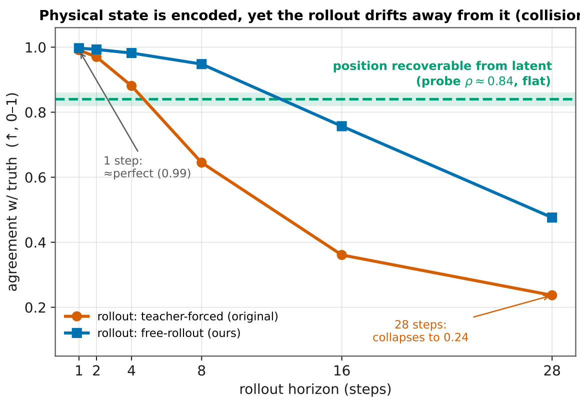

Figure 3: Decodable, but not load-bearing (collision domain). The flat dashed line is position decodability from the frame embeddings (probe $\rho \approx 0.84$): the state is *present* at every horizon. Yet the teacher-forced rollout (LeWM's default) collapses from near-perfect one-step agreement (0.99) to 0.24 by horizon 28; training the same model by free rollout (§3.4) stabilizes the long horizon (0.48) with no physics added. Presence of the state does not imply its use by prediction.

(supplementary)). Second, *realistic simulation*: trained directly on Physion++, every injection variant we ported (fixed slot, consistency, consistency+acceleration) degrades per-scenario rollout nMSE by 3–10× against the identically trained free-rollout model (Appendix B, supplementary); the gaps dwarf the three-seed noise band of the baseline. The probe family has not yet been run on Physion++, so this corroboration currently covers the slot and consistency families.

4.3 Mechanism: Decodable but Not Load-Bearing

The puzzle of §1 localizes here: position is linearly decodable from the latent at every horizon, yet the rollout drifts away from it (Figure 3)—*decodable, but not load-bearing*. The same dissociation is known for language models—probe-recoverable information need not be information the model uses (Elazar et al. 2021); the three tests explain why it arises here.

Test 1: the physical constraint is injected. Probing *only the* 190 *non-slot dimensions* decodes position at $\rho = 0.79$–0.92 across all three domains—as well as the full latent, undiminished by pinning the slot—while a random-2-dimension control fails (0.2–0.5). Position is a redundant, distributed copy the predictor can route through while ignoring any structured slot.

Test 2: the constraint acts harmfully. Both measurements move, which indicates the constraint acts. At probe weight λ=10 the weighted physical loss grows to 15–125× the prediction loss (the gradient-magnitude proxy), and the encoder's effective dimensionality collapses by 39–90%; at λ=1, the setting of §4.2, the ratio is milder (3–20×) yet the harm persists. With slot decodability rising $0.31 \to 0.96$, the constraint demonstrably reshapes the representation—in a direction that competes with prediction.

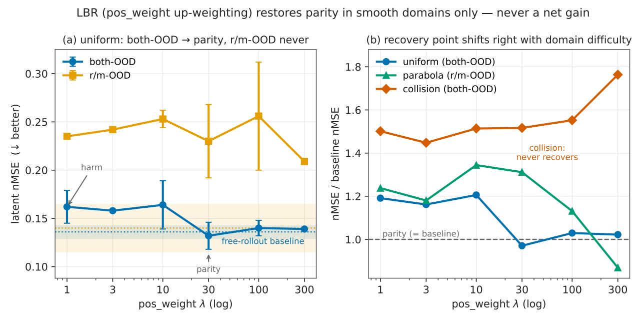

Figure 4: Dose–response re-weighting. Left: uniform, slot weight w=1 → 300 (the axis label pos_weight λ denotes w) against the baseline band—both-OOD returns to parity, r/m-OOD never does. Right: nMSE ratio to baseline across domains—collision stays above baseline at every weight and degrades as weight grows. Re-weighting validates the mechanism's direction but is not a repair.

Test 3: re-weighting the physical loss does not help. Sweeping the slot weight w=1 → 300 (Figure 4) moves the harm systematically—the mechanism's direction is confirmed—but weighting rescues only 2 of 4 domain–partition decision cells, to *parity* and never a net gain: uniform both-OOD returns exactly to baseline (0.132 ± 0.017 vs. 0.136 ± 0.008), uniform r/m-OOD never recovers, and collision worsens monotonically with weight. Insufficient physical loss is therefore not the root cause: the black-box route stays open at every weight, and closing it requires changing the architecture.

4.4 Free Rollout Reduces Error by 2.2–8.3×

Replacing teacher forcing with free rollout cuts the hardest-split error 2.2× (uniform, $0.300{\pm}0.007 \to 0.136{\pm}0.008$), 3.6× (parabola, $0.443{\pm}0.048 \to 0.122{\pm}0.007$), and 2.4× (collision, $1.152{\pm}0.048 \to 0.479{\pm}0.079$), with improvements on *every* partition including ID (2.0–4.6×)—it is not an OOD patch (Figure 5). On Physion++ the same switch is worth 8.3× at horizon 64 ($1.174{\pm}0.041 \to 0.141{\pm}0.018$, three seeds), and the entire pattern replicates on the frozen-DINOv2 backbone (Figure 5, right; §4.5). The horizon must match dynamics complexity: collision improves markedly at $H{\approx}20$ ($0.393 \to 0.294$) while smooth domains degrade with excess horizon (uniform horizon-28 cosine $0.969 \to 0.814$ at H=16); realistic-simulation scenes have the largest appetite—extending $H{=}8 \to 28$ (with geometric scale jitter stabilizing amplitude; Appendix D (supplementary)) drives horizon-64 nMSE from 0.280 to 0.014, a $\sim$19× reduction with no inflection found (Appendix D, supplementary).

As the positive control of §3.4, this result carries the methodological weight of the paper: on the same code, metrics, seeds, and data that judged all thirty injection cells, a physics-free intervention produces multiples—so the injection failure of §4.2 is attributable to the injected structure, not to the pipeline or the yardstick.

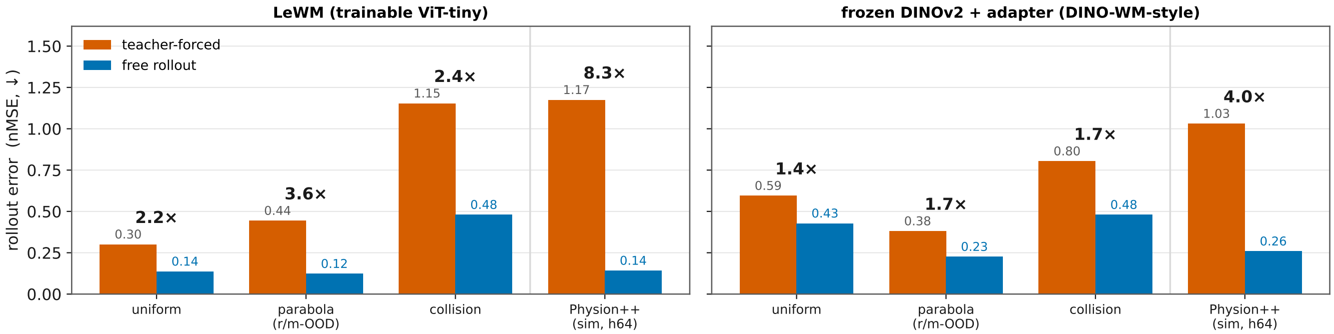

Figure 5: Teacher forcing vs. free rollout on both backbones (hardest clean split, nMSE ↓; fold-change above bars; all three-seed with non-overlapping intervals). Left, LeWM: uniform $0.300 \to 0.136$ (2.2×), parabola $0.443 \to 0.122$ (3.6×), collision $1.153 \to 0.479$ (2.4×), Physion++ horizon-64 $1.174 \to 0.141$ (8.3×). Right, the frozen-DINOv2 variant (§3.5): the same physics-free switch yields 1.4×/1.7×/1.7×/4.0×—the protocol lever is backbone-independent.

4.5 External Control: Extrinsic Is Necessary, Not Sufficient

Extrinsic port. The faithful extrinsic port (PIWM; §3.5) learns the correct physics and, where its encoder is in-distribution, beats our shared-latent model: rolled-out position ρ reaches 0.96 (ID) and 0.97 (v-OOD) vs. LeWM's 0.93/0.87. But under radius/mass shift—which changes appearance, hitting the VAE encoder—it collapses to $\rho = 0.33$ (r/m-OOD) and 0.48 (both-OOD) while LeWM holds 0.89/0.87. The architectural prediction of §4.3 is confirmed from the outside: making the physical state load-bearing by construction is what extrinsic designs buy (their equations alone do not transfer, §4.2); yet closing the bypass leaves the encoder-OOD problem unsolved. Extrinsic structure is necessary but not sufficient.

Cross-backbone replication. On the frozen-DINOv2 variant every headline finding replicates across three seeds. Presence is even stronger—the physics-naive frozen encoder decodes position at ρ=0.95 (vs. 0.90 for LeWM), with the same black-box bypass—so presence and bypass are properties of generic visual features, not our training. The protocol lever transfers (1.4–1.7× synthetic, 4.0× Physion++; Figure 5, right). Injection still never wins (27/30 at or below baseline; Appendix C (supplementary)); the three above-parity cells dissolve under content-free controls—a shuffled target, or a 10×-weaker pin at matched loss weighting, keeps the apparent gain while ID error degrades, i.e. capacity-restriction regularization, not physics (Trap 5, Appendix E (supplementary)). One difference: with the encoder frozen, injection is inert rather than harmful—the losses reshape only the adapter, bounding damage but foreclosing benefit. Across every control, the same sentence survives: physical state in a latent world model is *decodable but not load-bearing*.

5 Conclusion

We asked whether injecting physics can make a latent world model physically consistent, and answered with a controlled anatomy: across ten variants, two training regimes, three synthetic domains, and two realistic-simulation benchmarks, injection fails in 29 of 30 cells—with the exact gravitational form, with perfect labels, and *more* when co-trained from scratch. The reason is that the physics is already there: the shared latent redundantly encodes the injected state ($\rho =$ 0.79–0.92 from the non-slot dimensions alone), so injection adds no signal while consuming capacity and gradient—*decodable but not load-bearing*. All three tests support this account, and its two predictions held: re-weighting removes the slot's harm but only to parity, and an extrinsic architecture that closes the bypass does help—until the OOD shift hits its encoder (ρ 0.33 vs. 0.89): necessary, not sufficient. What actually moved prediction was never the structure: repairing the training protocol—free rollout, horizons matched to dynamics complexity—delivered 2.2–8.3× on identical code and metrics.

The constructive reading: to help a latent world model, an inductive bias must supply what the latent *lacks*—not its state content, but its long-horizon dynamics. Injections that target dynamics or conserved quantities rather than state, and extrinsic designs paired with OOD-robust encoders, are the two routes our results leave open—the former untested here beyond the suggestive velocity-on-parabola signature, the latter necessary but demonstrated insufficient alone. Our own evolution-targeting variants are deliberately simple instantiations spanning the design axes: their failure shows that pinning the evolution is not *automatically* useful, not that a stronger, dynamics-matched design could not work.

Limitations. Our training-dynamics conclusions rest primarily on one compact backbone (∼10M ViT-tiny JEPA); the headline findings replicate on a frozen-DINOv2 backbone (§4.5), where injection is inert rather than harmful—so the *harm* claim is established on the trainable-encoder regime. Headline comparisons, re-weighting, and the seed-refuted cells are three-seed; the remaining 26 scan cells and the Physion++ variants are single-seed, with effects 5–20 seed-std and 3–10× clear of the noise band. The probe family remains untested on Physion++, and the one favorable cell (velocity on parabola) suggests dynamics-matched injection deserves study we did not perform.

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